SCHEMES ARE SOBER

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2154-2166, following the statement at 2146-2152.

Editorial note. The uniqueness step 'it follows immediately' is the standard fact that a nonempty open subset of an irreducible closed set contains every generic point, plus uniqueness of generic points of V(p) in Spec(R).

Allowed standard prerequisites (editorial).

- Generic points in spectra of rings. Topology, Definition topology-definition-generic-point (cross-chapter, cited at ~2151); the affine fact V(p) has generic point p. Supplies the affine generic point.

Source reference. Original reference 004X: focused-seeds/geometry/S9-schemes-scheme-sober/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. Any irreducible closed subset of X has a unique generic point. In other words, X is a sober topological space, see Topology, Definition 004X.*

Proof. Let $Z \subset X$ be an irreducible closed subset. For every affine open $U \subset X$, $U = \mathrm{Spec}(R)$ we know that $Z \cap U = V(I)$ for a unique radical ideal $I \subset R$. Note that $Z \cap U$ is either empty or irreducible. In the second case (which occurs for at least one U) we see that $I = \mathfrak{p}$ is a prime ideal, which is a generic point ξ of $Z \cap U$. It follows that $Z = \overline{\{\xi\}}$, in other words ξ is a generic point of Z. If ξ' was a second generic point, then $\xi' \in Z \cap U$ and it follows immediately that $\xi' = \xi$. $\square$

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